

Dynamical Systems of Particles Around a Kerr Black Hole

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1 Introduction

Black holes, as predicted by general relativity, are among the most intriguing objects in our universe. In this report, we extend the non-rotating black holes to rotating black holes described by the Kerr metric. In 1963, Roy Kerr discovered an exact solution to Einstein's field equations describing a rotating, uncharged black hole. This solution has been extensively investigated due to its relevance in modeling astrophysical black holes, which are generally expected to have significant spin. A foundational overview of general relativity and its mathematical structure is provided in texts such as *A First Course in General Relativity* (2) and Wald's *General Relativity* (3).

This report investigates the dynamics of particles orbiting a Kerr black hole. By employing Hamiltonian and Hamilton–Jacobi formalisms, we derive key conserved quantities and analyze the motion of test particles in curved spacetime. Numerical simulations further illustrate complex phenomena, offering insights into high-energy astrophysical processes.

2 Overview of Kerr Black Hole

The spacetime geometry around a rotating black hole is described by the Kerr metric. In this study, we adopt the Boyer–Lindquist coordinates (t, r, θ, ϕ) , which conveniently represent the stationary and axisymmetric properties of Kerr spacetime. The metric components and detailed descriptions can be found in *A First Course in General Relativity* (2). For simplicity, we will set $G = c = 1$.

$$ds^2 = -\frac{\Delta - a^2 \sin^2 \theta}{\Sigma} dt^2 - 2a \sin^2 \theta \frac{r^2 + a^2 - \Delta}{\Sigma} dt d\phi \\ + \frac{\Sigma}{\Delta} dr^2 + \Sigma d\theta^2 + \frac{(r^2 + a^2)^2 - \Delta a^2 \sin^2 \theta}{\Sigma} \sin^2 \theta d\phi^2 \quad (1)$$

where Σ and Δ are defined as

$$\Sigma = r^2 + a^2 \cos^2 \theta \quad (2)$$

$$\Delta = r^2 + a^2 - 2Mr \quad (3)$$

The black hole's spin (or rotation) parameter is defined as $a = \frac{J}{M}$ where J is the total angular momentum of the black hole, and M is the black hole mass in geometrized units ($c = G = 1$). In

these units, a has the dimension of length. If $a = 0$, the black hole is nonrotating (a Schwarzschild black hole). When $a \neq 0$, the black hole is rotating.

We already know that a non-rotating black hole has a single event horizon, leading to a simple overall structure. But for a Kerr black hole, there is a unique region called the ergosphere, within which spacetime is dragged by the black hole's rotation so that no object can remain stationary relative to a distant observer.

$$r_{\text{ergosphere}} = M + \sqrt{M^2 - a^2 \cos^2 \theta} \quad (4)$$

Furthermore, the Kerr black hole possesses two horizons (an outer event horizon and an inner Cauchy horizon)

$$r_+ = M + \sqrt{M^2 - a^2} \quad (5)$$

$$r_- = M - \sqrt{M^2 - a^2} \quad (6)$$

and a ring-shaped singularity, making its internal configuration more intricate than that of a non-rotating black hole, which has only one horizon and a point-like singularity.

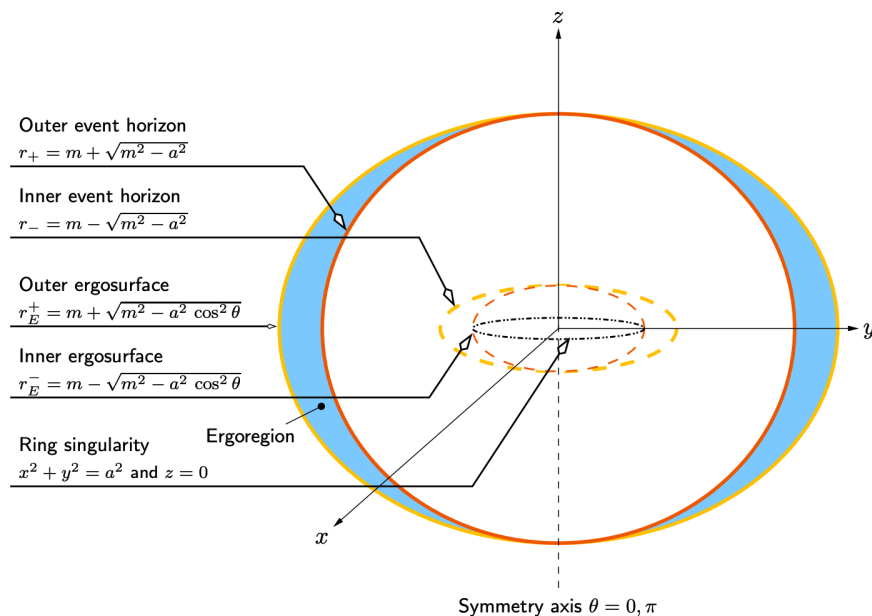


Figure 1: Schematic Kerr Black Hole (4)

In terms of symmetry, the Kerr solution is only axisymmetric because of its spin, whereas the non-rotating (Schwarzschild) solution is fully spherically symmetric. Overall, the rotation of the Kerr black hole introduces additional complexities such as frame dragging and the ergosphere, while the non-rotating black hole remains simpler and more symmetric. Notably, when the Kerr black hole's spin parameter approaches zero, its description reduces to the Schwarzschild solution.

3 Mathematical Description of the Dynamical System

We assumed the Kerr black hole to be stationary, so the metric is independent of the time, and we have a Killing vector $k = (1, 0, 0, 0)$. Also, we assume the black hole to be axisymmetric; without loss of generality, we assume it rotates around the z-axis. So we have another Killing vector $m = (0, 0, 0, 1)$. As always, for a geodesic with affine parameter λ , we use x to denote the position of the geodesic, just keep in mind that we are working with Boyer-Lindquist coordinates. We use u to denote the velocity vector.

Definition 3.1. *We first define the Lagrangian to be*

$$\mathcal{L}(x, \dot{x}) := \frac{1}{2}g(\dot{x}, \dot{x}) = \frac{1}{2}g_{\mu\nu}\dot{x}^\mu \dot{x}^\nu \quad (7)$$

Also, we define the conjugate momentum p_μ as

$$p_\mu := \frac{\partial \mathcal{L}}{\partial \dot{x}^\mu} = g_{\mu\nu} \dot{x}^\nu, \quad (8)$$

Notice that the physical momentum $p^\mu = u^\mu$, so lowering the index we have $p_\mu = g_{\mu\nu}u^\nu$ which is just the conjugate momentum defined above.

Lemma 3.1. *p_t and p_ϕ are constant along the geodesics.*

Proof. Let ξ denotes any of the two Killing vectors above. By the Killing vector equation

$$\nabla_\mu \xi_\nu + \nabla_\nu \xi_\mu = 0 \quad (9)$$

Since $u = p$, and by the geodesic equation

$$p^\nu \nabla_\nu p_\mu = u^\nu \nabla_\nu u_\mu = 0 \quad (10)$$

So consider how $\xi_\mu p^\mu$ change along the geodesic, we have

$$p^\nu \nabla_\nu (\xi_\mu p^\mu) = p^\nu [(\nabla_\nu \xi_\mu) p^\mu + \xi_\mu (\nabla_\nu p^\mu)] \quad (11)$$

$$= p^\nu (\nabla_\nu \xi_\mu) p^\mu \quad (12)$$

$$= \frac{1}{2} p^\nu p^\mu (\nabla_\nu \xi_\mu) + \frac{1}{2} p^\nu p^\mu (-\nabla_\mu \xi_\nu) \quad (13)$$

$$= 0 \quad (14)$$

□

So the geodesic motion in Kerr geometry gives us two constants p_t and p_ϕ . The energy and angular momentum are $E = -k^\mu u_\mu$ and $L = m^\mu u_\mu$. Hence, it is easy to see that $E = -p_t$ and $L = p_\phi$ are constant along the geodesic. We should not forget there is another trivial invariant κ which is the invariant mass of the test particle, defined by

$$\kappa = g_{\mu\nu} u^\mu u^\nu = g_{\mu\nu} p^\mu p^\nu \quad (15)$$

$$\kappa = \begin{cases} -1 & \text{for timelike geodesics,} \\ +1 & \text{for spacelike geodesics,} \\ 0 & \text{for null geodesics.} \end{cases}$$

In this section, we are going to show that there is another constant Q by using the Hamilton-Jacobi theory, and it is originally proved by Brandon Carter in his paper *Global Structure of the Kerr Family of Gravitational Fields* (5)

Definition 3.2. We define

$$Q := p_\theta^2 - \cos^2 \theta \left[(\kappa + E^2) a^2 - \frac{L^2}{\sin^2 \theta} \right] \quad (16)$$

Definition 3.3. We define the Hamiltonian to be a function of the coordinate and the conjugate momentum:

$$H(x^0, \dots, x^3, p_0, \dots, p_3) := p_\mu \dot{x}^\mu - \mathcal{L}(x, \dot{x}) \quad (17)$$

$$= p_\mu g^{\mu\nu} p_\nu - \frac{1}{2} g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu \quad (18)$$

$$= p_\mu g^{\mu\nu} p_\nu - \frac{1}{2} g_{\mu\nu} g^{\mu\alpha} p_\alpha g^{\nu\beta} p_\beta \quad (19)$$

$$= \frac{1}{2} p_\mu g^{\mu\nu} p_\nu \quad (20)$$

By easy computation, we have the Hamiltonian equations

$$\dot{x}^\mu = \frac{\partial H}{\partial p_\mu} \quad (21)$$

$$\dot{p}_\mu = - \frac{\partial H}{\partial x^\mu} \quad (22)$$

But solving this system of equations is not an easy job. So as mentioned above, here we used the Hamiltonian-Jacobi approach.

Theorem 3.1. Consider the Hamiltonian-Jacobi equation

$$H\left(x, \frac{\partial S}{\partial x}\right) + \frac{\partial S}{\partial \lambda} = 0 \quad (23)$$

where $\frac{\partial S}{\partial x} = (\frac{\partial S}{\partial x^0}, \dots, \frac{\partial S}{\partial x^3})$. If $S(x(\lambda), \lambda)$ solves the Hamiltonian-Jacobi equation, then

$$\frac{\partial S}{\partial x^\mu} = p_\mu \quad (24)$$

$$\frac{\partial S}{\partial \lambda} = -H \quad (25)$$

Proof. Similar to classical mechanics, see *Classical Mechanics, Chapter 10* (6), the solution S has the expression

$$S = \int \mathcal{L} d\lambda + \text{constant} \quad (26)$$

For simplicity, we just consider S to be the action. The variation in S is

$$\delta S = \int_{\lambda_0}^{\lambda} \left(\frac{\partial \mathcal{L}}{\partial x^\mu} \delta x^\mu + \frac{\partial \mathcal{L}}{\partial \dot{x}^\mu} \delta \dot{x}^\mu \right) d\tau \quad (27)$$

The trajectories obey the Euler-Lagrange equation and we must have

$$\frac{\partial \mathcal{L}}{\partial x^\mu} = \frac{d}{d\lambda} \frac{\partial \mathcal{L}}{\partial \dot{x}^\mu} \quad (28)$$

on the other hand $\delta \dot{x} = \frac{d}{d\lambda}(\delta x)$. Using these relations, we can rewrite the variation of S as

$$\delta S = \int_{\lambda_0}^{\lambda} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}} \frac{d(\delta x)}{d\lambda} + \frac{d}{d\lambda} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}} \right) \delta x \right) d\tau \quad (29)$$

$$= \int_{\lambda_0}^{\lambda} \frac{d}{d\tau} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}} \delta x \right) d\tau \quad (30)$$

$$= \left[\frac{\partial \mathcal{L}}{\partial \dot{x}} \delta x \right]_{\lambda_0}^{\lambda} \quad (31)$$

$$= p_\mu(\lambda) \delta x^\mu(\lambda) - p_\mu(\lambda_0) \delta x^\mu(\lambda_0) \quad (32)$$

Usually, $\delta x^\mu(\lambda_0) = 0$ because the initial endpoint is fixed. Hence,

$$\delta S = p_\mu(\lambda) \delta x^\mu(\lambda) \quad (33)$$

Comparing the coefficients of $\delta x^\mu(\lambda)$ on both sides, we obtain

$$\frac{\partial S}{\partial x^\mu} = p_\mu \quad (34)$$

Now consider letting the original trajectory continue along its optimal path. Then $dS = \mathcal{L} d\tau$. The total derivative of $S(x^\mu(\lambda), \lambda)$ as λ changes along an actual trajectory $x^\mu(\lambda)$:

$$\frac{dS}{d\lambda} = \frac{\partial S}{\partial x^\mu} \dot{x}^\mu + \frac{\partial S}{\partial \lambda} = p_\mu \dot{x}^\mu + \frac{\partial S}{\partial \lambda} = H + \mathcal{L} + \frac{\partial S}{\partial \lambda} \quad (35)$$

Hence we also have

$$\frac{\partial S}{\partial \lambda} = -H \quad (36)$$

□

Therefore, if we have the solution of S , then the expression of the conjugate moments p_μ and hence the expression of $\dot{x}^\mu$ can be solved. Then we can write the solution of geodesic equations in a closed form.

Corollary 3.1.1. *The solution of the Hamiltonian-Jacobi equation must have the form*

$$S = -\frac{1}{2} \kappa \lambda - E t + L \phi + S^{(r\theta)}(r, \theta) \quad (37)$$

where $S^{(r\theta)}$ is a function of r and θ to be determined.

Proof. By the theorem above and the previous discussion,

$$\frac{\partial S}{\partial \lambda} = -H = \frac{1}{2} g^{\mu\nu} p_\mu p_\nu = \frac{1}{2} \kappa \quad (38)$$

$$\frac{\partial S}{\partial t} = p_t = -E \quad (39)$$

$$\frac{\partial S}{\partial \phi} = p_\phi = -L \quad (40)$$

All of the quantities above are constant along the geodesic. $\square$

Theorem 3.2. *The quantity Q we defined is also constant along geodesics, it is called the Carter constant.*

Proof. We follow the idea of B. Carter (5), we want to find a separable solution, so we further assume S has the following expression

$$S = -\frac{1}{2}\kappa\lambda - Et + L\phi + S^{(r)}(r) + S^{(\theta)}(\theta) \quad (41)$$

We now want to substitute this expression into the Hamiltonian-Jacobi equation. In order to do this, we first compute that the inverse of the Kerr metric is

$$g^{-1} = \begin{pmatrix} -\frac{1}{\Delta}[r^2 + a^2] + \frac{2Mr a^2 \sin^2 \theta}{\Delta \Sigma} & 0 & 0 & -\frac{2Mr a}{\Sigma \Delta} \\ 0 & \frac{\Delta}{\Sigma} & 0 & 0 \\ 0 & 0 & \frac{1}{\Sigma} & 0 \\ -\frac{2Mr a}{\Sigma \Delta} & 0 & 0 & \frac{\Delta - a^2 \sin^2 \theta}{\Sigma \Delta \sin^2 \theta} \end{pmatrix} \quad (42)$$

Also by the expression $H = \frac{1}{2}p_\mu g^{\mu\nu} p_\nu$, the H-J equation reduces to

$$\frac{1}{2} \frac{\partial S}{\partial x^\mu} g^{\mu\nu} \frac{\partial S}{\partial x^\nu} + \frac{\partial S}{\partial \lambda} = 0 \quad (43)$$

$$-\kappa + \frac{\Delta}{\Sigma} \left(\frac{dS^{(r)}}{dr} \right)^2 + \frac{1}{\Sigma} \left(\frac{dS^{(\theta)}}{d\theta} \right)^2 - \frac{1}{\Delta} \left[r^2 + a^2 + \frac{2Mr^2 \sin^2 \theta}{\Sigma} \right] E^2 + \frac{4Mr a}{\Sigma \Delta} EL + \frac{\Delta - a^2 \sin^2 \theta}{\Sigma \Delta \sin^2 \theta} L^2 = 0 \quad (44)$$

Notice that

$$(r^2 + a^2) + \frac{2Mr a^2 \sin^2 \theta}{\Sigma} = \frac{1}{\Sigma} [(r^2 + a^2 \cos^2 \theta)(r^2 + a^2) + 2Mr a^2 \sin^2 \theta] \quad (45)$$

$$= \frac{1}{\Sigma} [(r^2 + a^2)^2 - (r^2 + a^2)^2 \sin^2 \theta + 2Mr a^2 \sin^2 \theta] \quad (46)$$

$$= \frac{1}{\Sigma} [(r^2 + a^2)^2 - a^2 \sin^2 \theta \Delta] \quad (47)$$

Substitute the equation above into the H-J equation and multiplying by Σ we have:

$$-\kappa \Sigma + \Delta \left(\frac{dS^{(r)}}{dr} \right)^2 + \left(\frac{dS^{(\theta)}}{d\theta} \right)^2 - \left[\frac{(r^2 + a^2)^2}{\Delta} - a^2 \sin^2 \theta \right] E^2 + \frac{4Mr a}{\Delta} EL + \left[\frac{1}{\sin^2 \theta} - \frac{a^2}{\Delta} \right] L^2 = 0 \quad (48)$$

The equation above only involves r and θ , so we rearrange the equation such that the LHS only involves r while the RHS only involves θ

$$\Delta \left(\frac{dS^{(r)}}{dr} \right)^2 - \kappa r^2 - \frac{(r^2 + a^2)^2}{\Delta} E^2 + \frac{4Mr a}{\Delta} EL - \frac{a^2 L^2}{\Delta} = - \left(\frac{dS^{(\theta)}}{d\theta} \right)^2 + \kappa a^2 \cos^2 \theta - a^2 \sin^2 \theta E^2 - \frac{1}{\sin^2 \theta} L^2 \quad (49)$$

By our definition of Q , we see that

$$-Q = -\left(\frac{\partial S}{\partial \theta}\right)^2 + \kappa a^2 \cos^2 \theta + a^2 \cos^2 \theta E^2 - \frac{L^2}{\sin^2 \theta} \cos^2 \theta \quad (50)$$

which is very similar to the RHS of the equation above. In order to make the RHS = $-Q$, we add $a^2 E^2 + L^2$ to both sides, so the equation becomes

$$\Delta \left(\frac{dS^{(r)}}{dr} \right)^2 - \kappa r^2 - \frac{(r^2 + a^2)^2}{\Delta} E^2 + \frac{4Mra}{\Delta} EL - \frac{a^2 L^2}{\Delta} + a^2 E^2 + L^2 = -Q \quad (51)$$

Since Q is independent of r , and by the expression above, Q is also independent of θ , therefore it is also a constant. $\square$

Theorem 3.3. *We define the functions $R(r)$ and $\Theta(\theta)$ as:*

$$R(r) = \Delta \left[Q + \kappa r^2 - (L - aE)^2 \right] + \left[E(r^2 + a^2) - La \right]^2 \quad (52)$$

$$\Theta(\theta) = Q + \cos^2 \theta \left[(\kappa + E^2) a^2 - \frac{L^2}{\sin^2 \theta} \right] \quad (53)$$

Then we have the following two ODEs:

$$p_\theta^2 = (\Sigma \dot{\theta})^2 = \Theta(\theta), \quad p_r^2 = \left(\frac{\Sigma}{\Delta} \dot{r} \right)^2 = \frac{R(r)}{\Delta^2} \quad (54)$$

Proof. $p_\theta^2 = \Theta(\theta)$ follows immediately after the expression of Q . Also, by the metric of Kerr spacetime

$$p_\theta = g_{\theta\nu} \dot{x}^\nu = g_{\theta\theta} \dot{\theta} = \Sigma \dot{\theta} \quad (55)$$

Now consider p_r , we already know that

$$\Delta \left(\frac{dS^{(r)}}{dr} \right)^2 - \kappa r^2 - \frac{(r^2 + a^2)^2}{\Delta} E^2 + \frac{4Mra}{\Delta} EL - \frac{a^2 L^2}{\Delta} + a^2 E^2 + L^2 = -Q \quad (56)$$

Notice that

$$\frac{4Mra}{\Delta} EL = \frac{2aLE}{\Delta} \cdot 2Mr = \frac{2aLE}{\Delta} (r^2 + a^2 - r^2 - a^2 + 2Mr) = \frac{2aLE}{\Delta} (r^2 + a^2 - \Delta) \quad (57)$$

Therefore, using this result and the definition of $R(r)$, we get

$$p_r^2 = \frac{R}{\Delta^2} \quad (58)$$

Similarly, using the metric of Kerr spacetime, we immediately have $p_r = \frac{\Sigma}{\Delta} \dot{r}$ $\square$

Theorem 3.4. *For the parameters ϕ , t of the geodesics, we have the following expression:*

$$\Sigma \dot{\phi} = -\left(aE - \frac{L}{\sin^2 \theta}\right) + \frac{a}{\Delta} [E(r^2 + a^2) - aL] \quad (59)$$

$$\Sigma \dot{t} = -(a^2 E \sin^2 \theta - aL) + \frac{r^2 + a^2}{\Delta} [E(r^2 + a^2) - aL] \quad (60)$$

Proof. By solving the system of ODEs below:

$$\begin{cases} p_t = g_{tt}\dot{t} + g_{t\phi}\dot{\phi} = -E \\ p_\phi = g_{\phi t}\dot{t} + g_{\phi\phi}\dot{\phi} = L \end{cases} \quad (61)$$

□

In summary, the geodesics of a test particle around the Kerr black hole can be expressed as:

$$\dot{\theta} = \pm \frac{1}{\Sigma} \sqrt{\Theta}, \quad \dot{r} = \pm \frac{1}{\Sigma} \sqrt{R} \quad (62)$$

$$\dot{\phi} = -\frac{1}{\Sigma} \left(aE - \frac{L}{\sin^2 \theta} \right) + \frac{a}{\Delta \Sigma} [E(r^2 + a^2) - aL] \quad (63)$$

$$\dot{t} = -\frac{1}{\Sigma} (a^2 E \sin^2 \theta - aL) + \frac{r^2 + a^2}{\Delta \Sigma} [E(r^2 + a^2) - aL] \quad (64)$$

4 Numerical Solutions and Visualizations

Theoretically we can using the integral to express them, but in reality, one can use numerical method to approximate the solution.

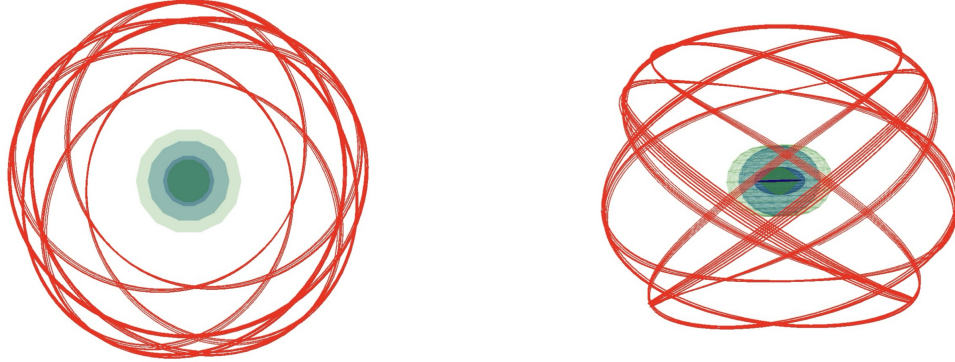


Figure 2: Geodesic of a test particle around Kerr black hole (7)

This image (2) shows a test particle's geodesic (the red curve) around a Kerr black hole with a spin parameter $a = 0.9$ and mass $M = 1$ in finite time. The initial values are set by $r_0 = 7$, $\theta_0 = 1.5707963267948966$, $\phi_0 = 0$, $\lambda_0 = 0$, $p_r = 0$ and $p_\theta = 1.9558$. The orbit is displayed in a 3D Cartesian coordinate system, illustrating the complex, precessing trajectory typical of Kerr spacetimes. The left subgraph represents the x-y plane while the right represents x-z plane.

5 Extension to Charged Particle Dynamics

In reality, we need to consider more complicated assumptions. For a black hole dipole magnetosphere, by solving equations of motion of a charged particle moving in the electromagnetic field numerically (8), we can see the gyration around the magnetic field line, the drift in the toroidal direction and the multi-periodic motion in the poloidal plane. Such a orbit is very complicated in both poloidal and toroidal directions. Although a stationary and axisymmetric magnetosphere is assumed, there are two conserved quantities; that is, the energy and angular momentum of the charged particle. In addition to these, the rest mass of the particle is the third invariable quantity. However, the fourth invariable quantity related to the azimuthal motion of the particle does not exist when the electromagnetic field is considered around a black hole. Thus, the motion of a test charge in a black hole magnetosphere is not integrable, and chaotic motion is expected. (8)

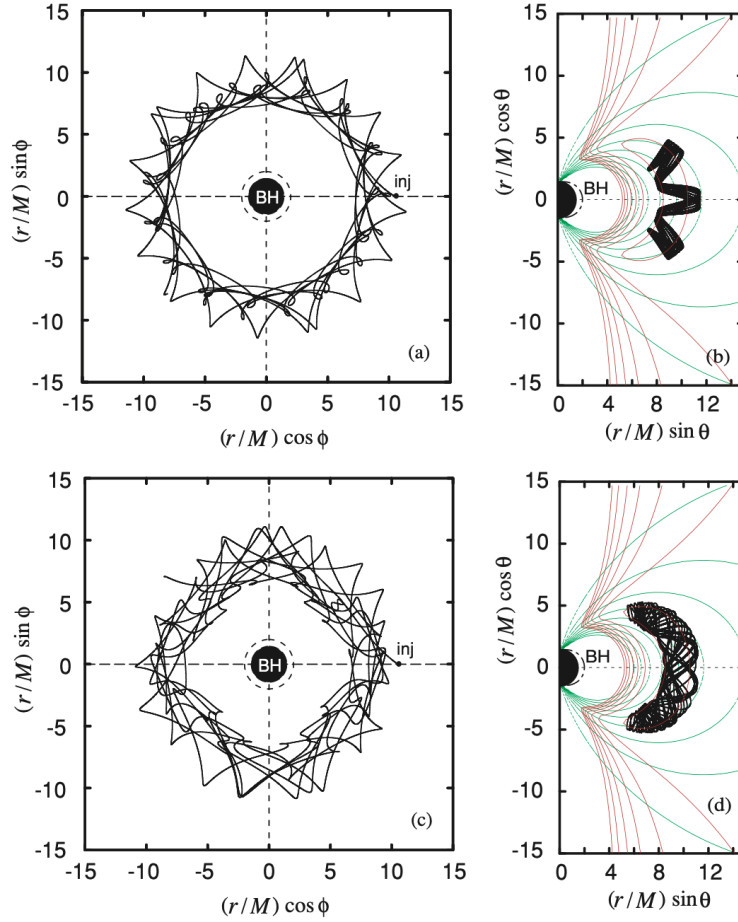


Figure 3: Examples of the trajectories of a test charge for $a = 0.9M$ (8)

Consider the image(3) from *Chaotic motion of Charged Particles in an Electromagnetic Field*

Surrounding a Rotating Black Hole (8). The left column shows the projections on the equatorial plane, and the right column shows the projections on the poloidal plane. Each sub-figure corresponds to different initial conditions or parameters. This shows that depending on initial conditions (e.g., injection angles, particle energies), the trajectories can become chaotic, filling regions of phase space in a complex, seemingly random manner. These chaotic orbits could have implications for high-energy emission, particle acceleration, and the overall dynamics of magnetized plasma in black-hole environments.

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